

# File 32: GT Intern Workstream — Page 1: Rosgen Stream Classification & Field Survey Processing

**Hydrologic Engineering & Geomorphic Design Manual Target School:** Georgia Institute of Technology (Georgia Tech) School of Civil & Environmental Engineering **Project Area:** Upper Savannah Basin & Chattahoochee River Spawning Headwaters **Supervising Board:** Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)

## 1. Scientific Principles: Natural Channel Design (NCD)

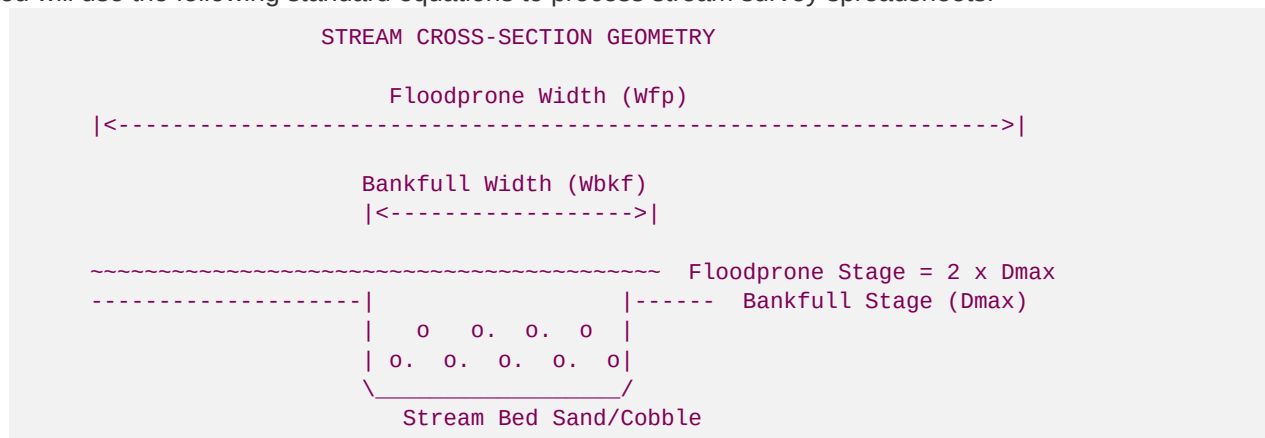
As a Geomorphology & Hydrologic Engineering Intern from Georgia Tech, you are responsible for translating physical stream surveys into stable, self-scouring channel geometries.

We utilize Dave Rosgen's **Natural Channel Design (NCD)** methodology. Degraded mountain streams are often in a state of high geomorphic instability, characterized by rapid lateral erosion (Type F or G channels) due to historic bank modifications. Our design goal is to transition these unstable reaches back to their stable historic geomorphic types (typically **Rosgen B or C channels** in montane ecoregions).

Your first task is to process field-collected cross-section, longitudinal profile, and substrate data to calculate the baseline and design geomorphic parameters.

## 2. Geomorphic Parameters and Core Equations

You will use the following standard equations to process stream survey spreadsheets:



### A. Entrenchment Ratio (ER)

The entrenchment ratio measures the degree of vertical containment of a stream channel within its valley, representing the stream's access to its active floodplain.

$$ER = \frac{W_{fp}}{W_{bkf}}$$

Where:

- $W_{bkf}$  = \text{Bankfull Width (width of channel at bankfull water stage)}.
- $W_{fp}$  = \text{Floodprone Width (width of valley at an elevation equal to twice the maximum bankfull depth, } 2 \cdot D\_{max}\text{)}.
- *Classification*: Stable Rosgen B and C channels require  $ER \geq 1.4$  (moderately to slightly entrenched). An  $ER < 1.4$  indicates severe channel incision (incapable of flooding onto the valley flats, resulting in high shear stress and bank collapse).

## B. Width-to-Depth Ratio (WDR)

The width-to-depth ratio indicates the channel's cross-sectional shape and is a primary indicator of sediment transport capacity.

$$WDR = \frac{W_{bkf}}{d_{mean}}$$

Where:

- $d_{mean}$  = \text{Mean bankfull depth } ( $A_{bkf} / W_{bkf}$ ), where  $A_{bkf}$  is the bankfull cross-sectional area.
- *Classification*: Stable high-gradient mountain streams (B-type) typically maintain a WDR between \$12 and \$20. An overwidened stream ( $WDR > 25$ ) experiences a drop in shear stress, causing heavy silt deposition.

## C. Sinuosity (K)

Sinuosity measures the degree of stream meandering.

$$K = \frac{L_{channel}}{L_{valley}}$$

Where:

- $L_{channel}$  = \text{Stream channel length measured along the thalweg (deepest point of flow)}.
- $L_{valley}$  = \text{Straight-line valley length between the upstream and downstream points}.

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# 3. Step-by-Step Geomorphic Survey Processing

You will compile field-collected total station or RTK GPS coordinates to classify the stream reach:

## Step 1: Process the Cross-Section Data

1. Open the survey CSV sheet containing *Station* ( $x$ ) and *Elevation* ( $y$ ) points across the channel cross-section.
2. Identify the **Bankfull Stage** elevation (marked by geomorphic indicators in the field, such as the top of flat depositional benches or changes in vegetation).
3. Calculate the bankfull area ( $A_{bkf}$ ), bankfull width ( $W_{bkf}$ ), and maximum depth ( $D_{max}$ ) relative to this elevation.
4. Calculate  $d_{mean} = A_{bkf} / W_{bkf}$ .

## Step 2: Process the Longitudinal Profile Data

1. Extract the elevation points along the channel thalweg, bankfull stage, and water surface over a length of at least 20 stream widths.
2. Calculate the average channel slope ( $S$ ):

$$S = \frac{\text{Thalweg Elevation}_{up} - \text{Thalweg Elevation}_{down}}{\text{Stream Length}}$$

3. Identify pool-to-pool spacing and riffle slopes.

### Step 3: Classify the Substrate ( $D_{50}$ )

1. Plot the Wolman Pebble Count data as a cumulative percent finer curve.
2. Extract the  $D_{50}$  value (the particle size at which 50% of the substrate is finer).
3. If  $D_{50} = 0.5 \text{ mm}$ , substrate is sand (Rosgen designation 5). If  $D_{50} = 22 \text{ mm}$ , substrate is gravel (designation 4). If  $D_{50} = 90 \text{ mm}$ , substrate is cobble (designation 3).

## 4. Geomorphic Rosgen Classification Lookup Matrix

Using the parameters calculated above, classify your stream reach according to the following lookup matrix:

| Stream Type | Entrenchment Ratio (ER)         | Width/Depth (WDR)  | Sinuosity (K) | Slope (S) | Geomorphic Description                                       |
|-------------|---------------------------------|--------------------|---------------|-----------|--------------------------------------------------------------|
| <b>A</b>    | < 1.4 (Highly Entrenched)       | < 12               | 1.0–1.2       | > 0.04    | Steep, entrenched, cascading step-pool stream.               |
| <b>B</b>    | 1.4–2.2 (Moderately Entrenched) | > 12               | > 1.2         | 0.02–0.04 | Stable, rapids and rapids-dominated steps, low erosion risk. |
| <b>C</b>    | > 2.2 (Slightly Entrenched)     | > 12               | > 1.2         | < 0.02    | Meandering pool-riffle channel with broad floodplain.        |
| <b>F</b>    | < 1.4 (Entrenched)              | > 12 (Very Wide)   | > 1.2         | < 0.02    | Highly unstable, entrenched, severely widening stream bend.  |
| <b>G</b>    | < 1.4 (Entrenched)              | < 12 (Very Narrow) | > 1.2         | 0.02–0.04 | Unstable "gully" channel with active severe bank slumping.   |

**Restoration Goal:** If a reach is classified as an eroding **F4** channel, your AutoCAD designs must narrow the bankfull channel and create a new floodplain bench to transition it to a stable **C4** or **B4** channel.